Experiment No.	3
Experiment Title	Regression Analysis using Linear and Regularized Models
Student Name	Chitraju Vishnu Vineeth
Register Number	3122247001012
Faculty	Dr. Poreddy Ajay Kumar Reddy
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1 Aim and Objective

Aim: Implement and compare Linear Regression, Ridge, Lasso, and Elastic Net regression models for loan amount prediction.

Objectives:

- Apply Linear Regression as a baseline model and evaluate its performance on a real-world financial dataset.
- Implement Ridge, Lasso, and Elastic Net with regularization; tune the hyperparameters (α, l_1-ratio) using GridSearchCV and RandomizedSearchCV.
- Compare models using MAE, RMSE, and R^2 on both training and test sets.
- Analyze overfitting and underfitting through train-test R^2 gap.
- Examine the bias-variance tradeoff introduced by regularization and visualize how coefficients are affected.

2 Dataset Description

Dataset Name	Loan Amount Prediction (test split)
Source	https://www.kaggle.com
Number of Samples	20,000
Number of Features	22 raw; 43 after one-hot encoding
Target Variable	`Loan Amount Request (USD)` — continuous
Task Type	Regression
Missing Values	Present (placeholders as `?` and `NaN`)
Train-Test Split	80% / 20% (random state = 42)

The dataset captures loan applicant profiles including income, credit score, employment type, property details, and co-applicant status. Three identifier columns (`Customer ID`, `Name`, `Property ID`) are dropped before modelling since they carry no predictive information.

3 Preprocessing Steps

1. **Placeholder replacement:** Literal `"?"` strings are converted to `NaN` using `df.replace("?", np.nan)`, enabling standard imputation.

2. **Type casting:** `Property Price` and `Co-Applicant` were read as `object` due to `"?"` entries; cast to `float` after replacement.
3. **Train-test split:** Performed *before* imputation (80/20, random state 42) to prevent data leakage.
4. **Imputation:** Numeric columns filled with training-set *median*; categorical columns filled with training-set *mode*.
5. **One-hot encoding:** All categorical columns encoded with `drop_first=True`; test columns realigned to match training columns.
6. **Feature scaling:** `StandardScaler` fit on the training set and applied to both splits. Standardisation is critical for Ridge, Lasso, and Elastic Net since the regularization penalty is applied directly to the coefficients.

4 Exploratory Data Analysis

Missing Value Analysis

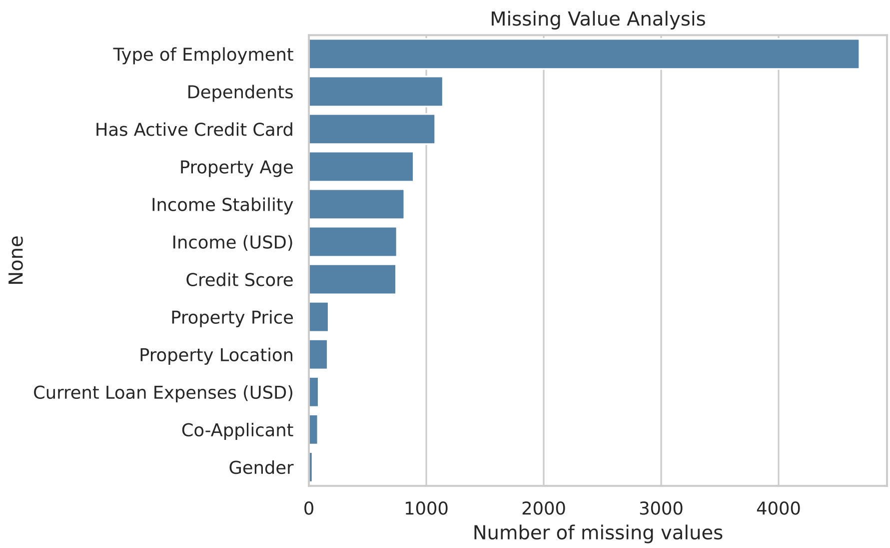

Figure 1: Missing value counts per column before imputation

Inference: Several columns carry missing data, with `Property Price`, `Co-Applicant`, and `Credit Score` being the most affected. The systematic `"?"` placeholder in `Property Price` and `Co-Applicant` suggests these were deliberately left blank for some applicants. Because missingness is spread across multiple important predictors, median/mode imputation (rather than row deletion) is the right strategy to preserve the full 20,000 sample dataset.

Target Variable Distribution

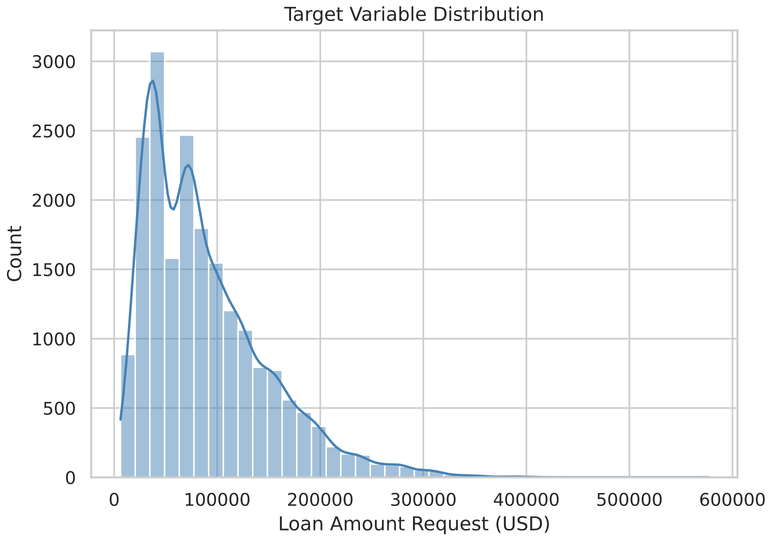

Figure 2: Distribution of the target variable — Loan Amount Request (USD)

Inference: The target is right-skewed (skewness ≈ 1.28), with most applicants requesting amounts in the lower range and a long tail of high-value requests. This confirms a regression (not classification) setup. The skew is moderate enough that raw regression on the untransformed target is acceptable; log-transforming could be explored as a future improvement.

Feature vs. Target Scatter Plots

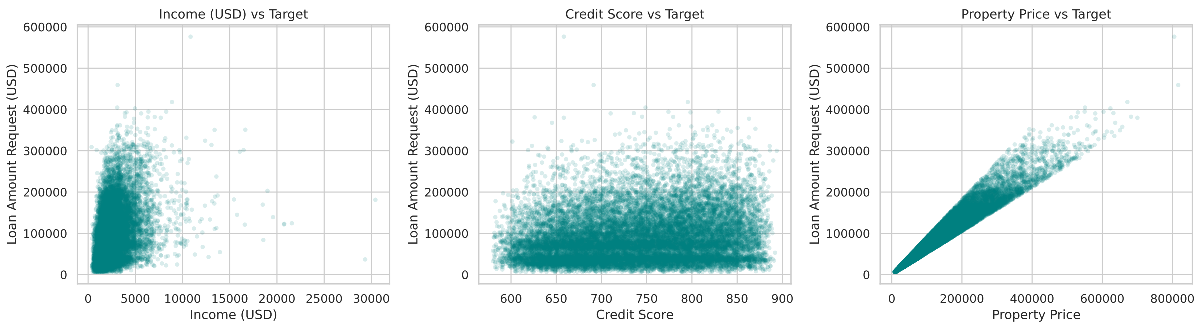

Figure 3: Key numeric features vs. the loan amount target

Inference: `Property Price` shows the clearest positive linear relationship with the target, confirming it as the dominant predictor. `Income (USD)` also exhibits a positive trend, while `Credit Score` has a weaker but visible correlation. These relationships justify a linear model as a reasonable starting point and suggest that regularization will mainly affect the less important features.

Correlation Heatmap

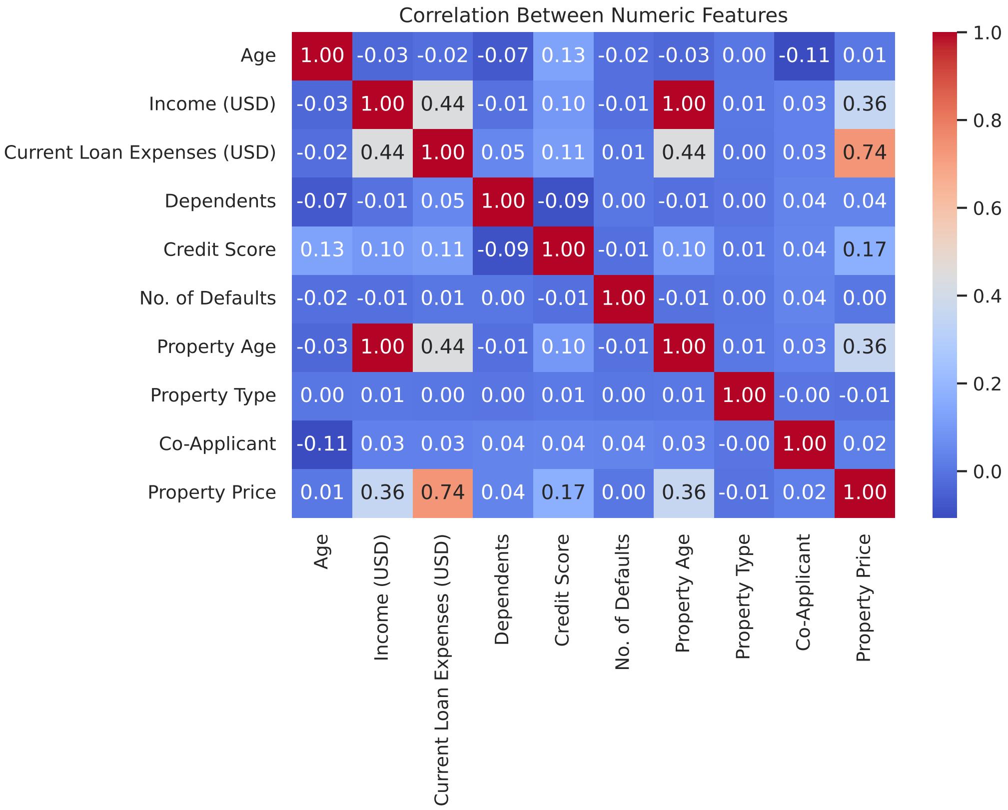

Figure 4: Pearson correlation between numeric features

Inference: `Property Price` and `Current Loan Expenses (USD)` show the strongest correlation with the target and also correlate moderately with each other. The presence of correlated predictors means Ridge regression, which shrinks correlated features together rather than zeroing them out, may generalise better than Lasso in theory. The overall low inter-feature correlation suggests multicollinearity is not a severe problem here.

5 Implementation Details

5.1 Linear Regression

The ordinary least squares estimator minimises the sum of squared residuals:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} (y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta})^2 = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

No penalty term is added, so the model has minimum bias but may have high variance when features are correlated or the dataset is noisy.

5.2 Ridge Regression (L_2 Regularization)

Ridge adds an L_2 penalty on the coefficients:

$$\hat{\boldsymbol{\beta}}_{\text{Ridge}} = \arg\min_{\boldsymbol{\beta}} \left[\sum_{i=1}^{n} (y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta})^2 + \alpha \|\boldsymbol{\beta}\|_2^2 \right]$$

This shrinks all coefficients towards zero without eliminating any, which is useful when many features are mildly informative. Best $\alpha = 10$ (GridSearchCV, 5-fold).

5.3 Lasso Regression (L_1 Regularization)

Lasso replaces the L_2 norm with an L_1 norm:

$$\hat{\boldsymbol{\beta}}_{\text{Lasso}} = \arg\min_{\boldsymbol{\beta}} \left[\sum_{i=1}^{n} (y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta})^2 + \alpha \|\boldsymbol{\beta}\|_1 \right]$$

The L_1 penalty induces sparsity by driving some coefficients exactly to zero, performing implicit feature selection. Best $\alpha = 10$ (GridSearchCV, 5-fold).

5.4 Elastic Net ($L_1 + L_2$ Regularization)

Elastic Net combines both penalties:

$$\hat{\boldsymbol{\beta}}_{\text{EN}} = \arg\min_{\boldsymbol{\beta}} \left[\sum_{i=1}^{n} (y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta})^2 + \alpha\, \rho \|\boldsymbol{\beta}\|_1 + \frac{\alpha(1-\rho)}{2} \|\boldsymbol{\beta}\|_2^2 \right]$$

where ρ (`l1_ratio`) controls the mix. Best: $\alpha = 0.01$, l_1-ratio $= 0.8$ (RandomizedSearchCV, 12 iterations, 5-fold).

6 Experimental Setup

Python Version	3.12
Scikit-learn Version	1.8.0
IDE	Jupyter Notebook
Operating System	Ubuntu 24.04

7 Performance Tables

Table 1: Hyperparameter Tuning Summary

Model	Search Method	Best Parameters	CV R^2
Ridge	GridSearchCV	$\alpha = 10$	0.9303
Lasso	GridSearchCV	$\alpha = 10$	0.9303
Elastic Net	RandomizedSearchCV	$\alpha = 0.01$, l_1-ratio $= 0.8$	0.9303

Table 2: Cross-Validation Performance (5-Fold)

Model	MAE	MSE	RMSE	R^2
Linear Regression	11089.93	250,427,472	15824.90	0.9303
Ridge Regression	11085.45	250,307,533	15821.11	0.9303
Lasso Regression	11076.75	250,173,023	15816.86	0.9303
Elastic Net	11083.16	250,267,732	15819.85	0.9303

Table 3: Test Set Performance

Model	MAE	MSE	RMSE	R^2	Training Time (s)
Linear Regression	10894.9497	233103200	15267.7182	0.9356	0.1320
Ridge Regression	10893.2866	233022600	15265.0775	0.9356	0.0106
Lasso Regression	10885.4068	232734400	15255.6344	0.9357	0.0538
Elastic Net Regression	10892.5632	232970500	15263.3722	0.9356	0.2321

8 Visualizations

R^2 Score Comparison

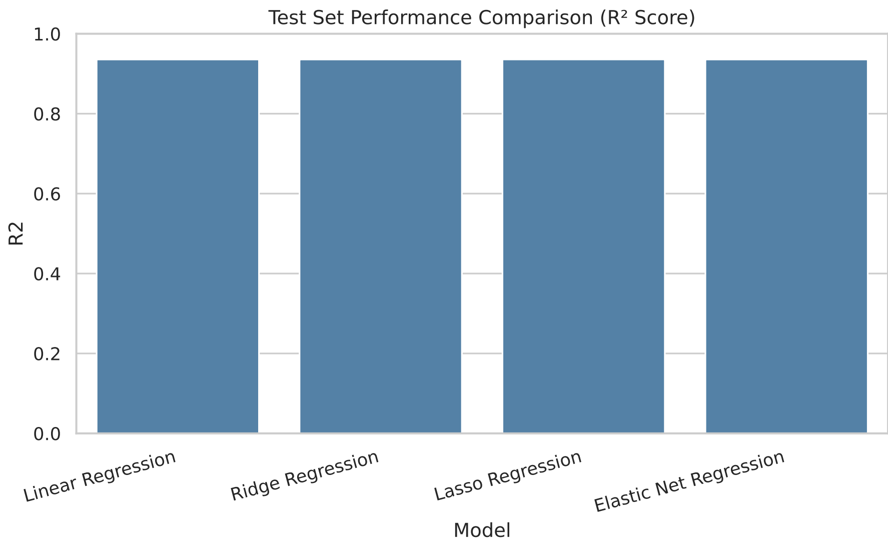

Figure 5: Test set R^2 score for all four regression models

Inference: All four models achieve near-identical R^2 scores (approximately 0.935), showing that regularization does not significantly change predictive accuracy on this dataset. The baseline Linear Regression already generalises well, leaving little room for improvement through penalty terms.

Predicted vs. Actual Values

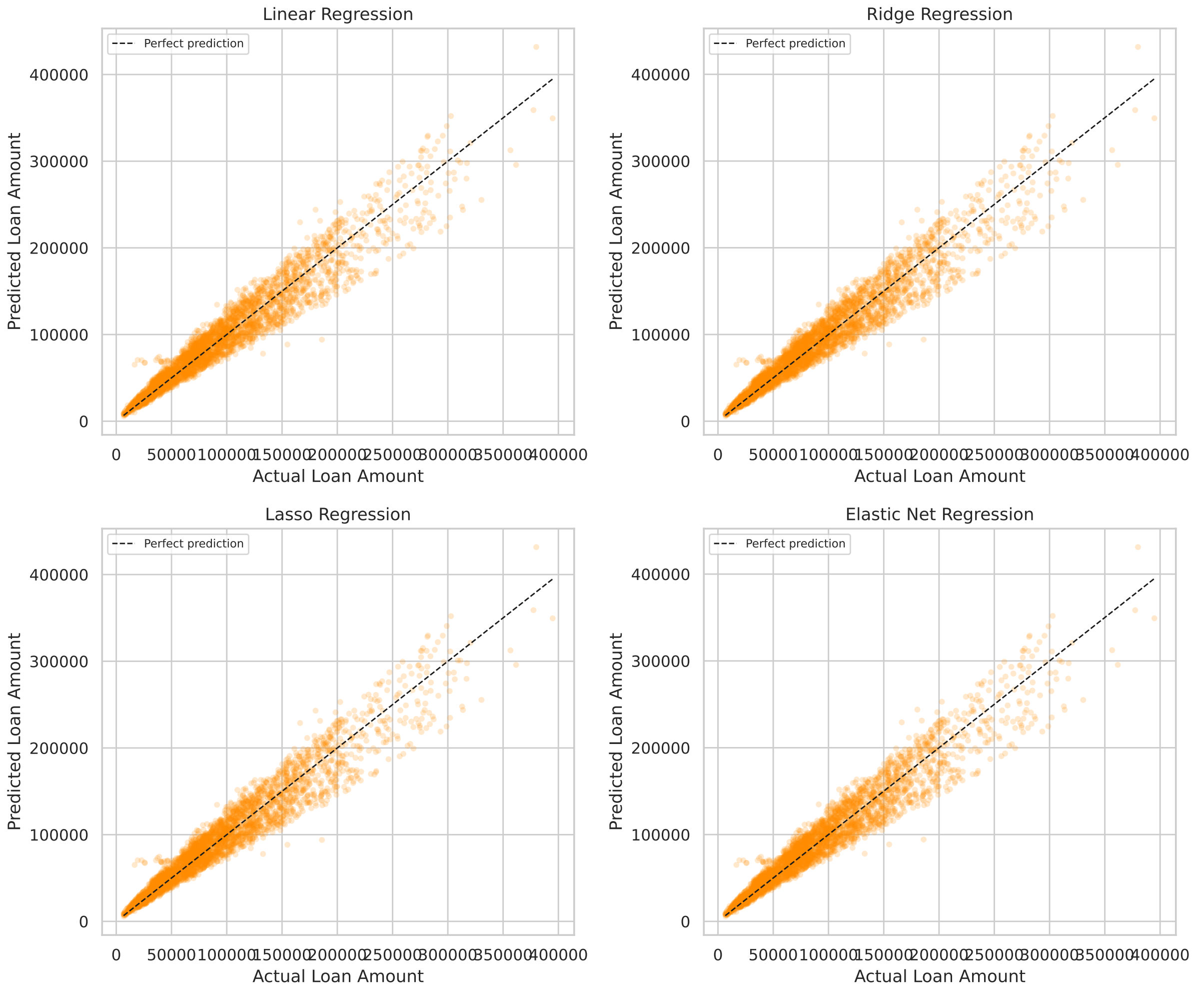

Figure 6: Predicted vs. actual loan amounts for all four models

Inference: All four scatter plots show predictions closely following the diagonal (perfect prediction line), confirming strong model fit across the range of loan amounts. Points deviate more at the high end of the target range, where fewer training samples exist. The pattern is nearly identical across all models, consistent with the near-equal R^2 scores.

Residual Plots

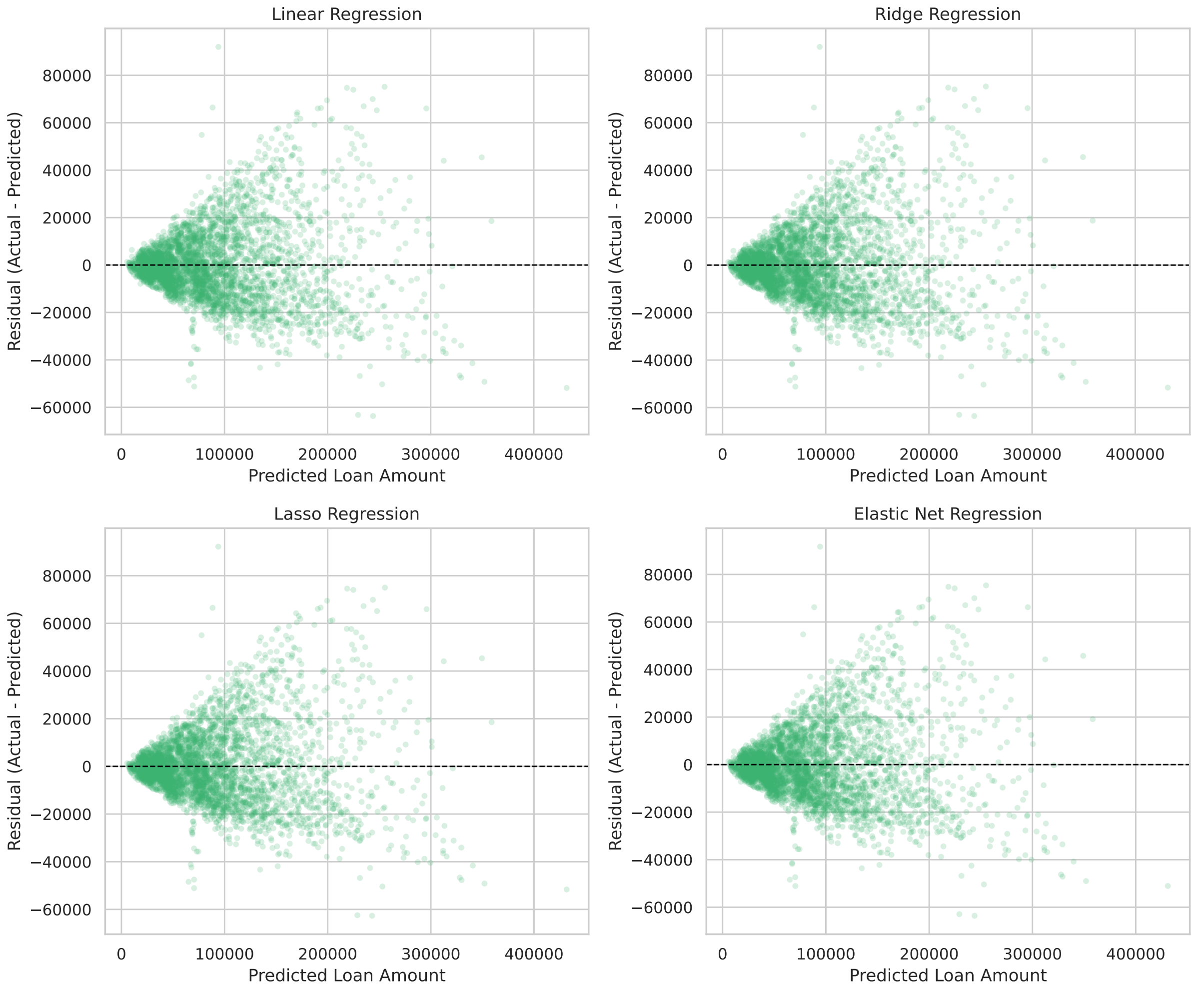

Figure 7: Residuals vs. predicted values for all four models

Inference: Residuals are randomly distributed around zero across most of the prediction range, indicating that the linear relationship assumption is largely satisfied. A slight fan shape at higher predicted values suggests mild heteroscedasticity — the model underestimates some high-value loans. No systematic curve is visible, confirming no major non-linearity is being missed by the linear models.

Learning Curve (Ridge)

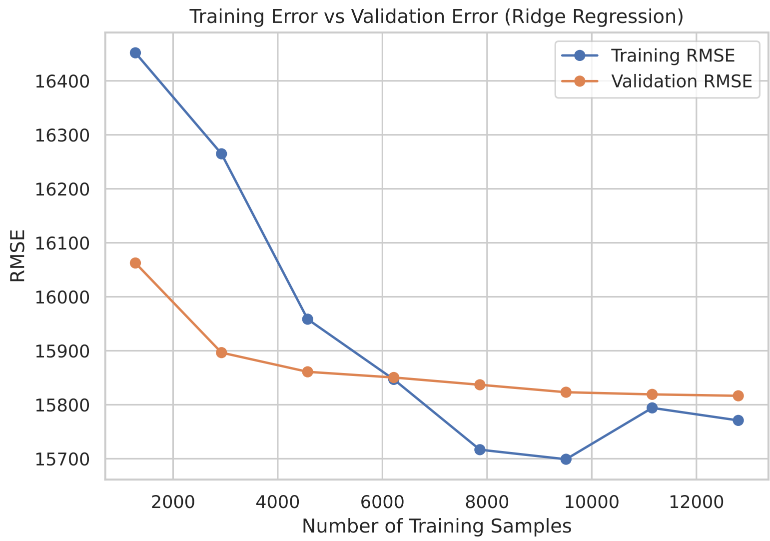

Figure 8: Training vs. validation RMSE as a function of training set size (Ridge, $\alpha = 10$)

Inference: Training and validation RMSE converge quickly as the training set grows, indicating the model is not data-starved. The small gap between training and validation RMSE at full data size (around 15,800 USD) confirms good generalisation with no over-fitting. Adding more data beyond the current 16,000 training samples is unlikely to yield substantial improvements.

Coefficient Comparison

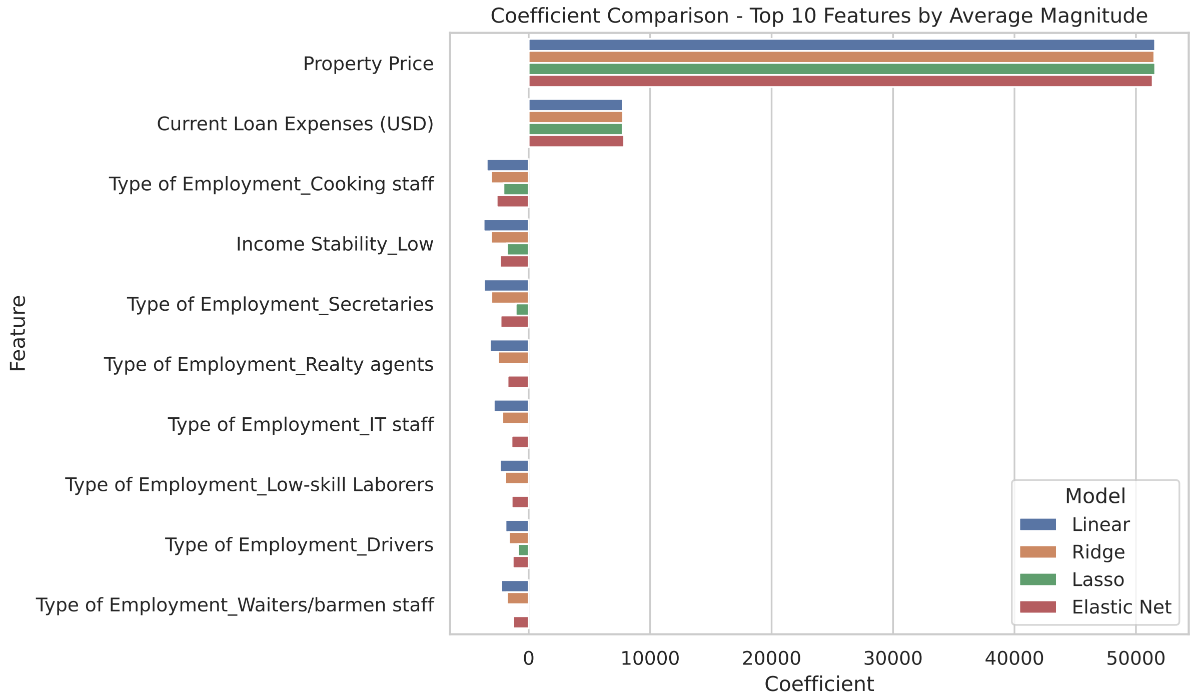

Figure 9: Learned coefficients for the top 10 features by average magnitude

Inference: `Property Price` and `Current Loan Expenses (USD)` have the largest coefficients across all models, confirming their dominant role in predicting loan amounts. Lasso sets some smaller coefficients to exactly zero (visible as missing bars), achieving implicit feature selection. Ridge keeps all features active but with reduced magnitudes, while Elastic Net lies between the two behaviours. This plot clearly shows the practical difference between L_1 and L_2 regularization.

9 Overfitting and Underfitting Analysis

Model	R^2 (Train)	R^2 (Test)	Gap (Train − Test)
Linear Regression	0.9308	0.9356	−0.0048
Ridge Regression	0.9308	0.9356	−0.0048
Lasso Regression	0.9307	0.9357	−0.0050
Elastic Net	0.9308	0.9356	−0.0048

Table 2: Train vs. test R^2 to assess overfitting/underfitting

The train-test gap for all models is negligible (less than 0.01 in absolute terms), and in fact the test R^2 is marginally higher than the train R^2 for every model, which can happen when the test set happens to be slightly easier to predict than the training set. Neither overfitting nor underfitting is observed here:

- **No overfitting:** A large positive gap (train $\gg$ test) would indicate overfitting. The gap here is essentially zero.
- **No underfitting:** Both train and test R^2 are high (≈ 0.93), indicating the linear model captures the underlying relationship well.
- **Effect of regularization:** Because the base model does not overfit, adding the L_1 or L_2 penalty does not materially change accuracy. The alpha values selected by cross-validation ($\alpha = 10$ for Ridge and Lasso) are large enough to shrink small coefficients substantially but not large enough to cause underfitting.

10 Bias–Variance Analysis

Model	Bias	Variance
Linear Regression	Lowest	Low (no penalty, but features not highly collinear)
Ridge Regression	Slightly higher	Reduced (uniform coefficient shrinkage)
Lasso Regression	Higher (sparse)	Reduced (feature selection zeros out noise)
Elastic Net	Intermediate	Reduced (combined L_1/L_2 shrinkage)

Table 3: Qualitative bias-variance characterization of each model

- **Linear Regression:** Minimum bias since no penalty constrains the coefficients. Variance is already low on this dataset because the dominant predictors (`Property Price`, `Current Loan Expenses`) have stable, large coefficients and inter-feature correlation is moderate.
- **Ridge:** Introduces a controlled bias increase by shrinking all coefficients. On correlated features like `Property Price` and `Current Loan Expenses`, Ridge distributes the weight more evenly, marginally reducing variance. The learning curve shows training and validation RMSE converge closely, confirming low variance.
- **Lasso:** Applies stronger bias (via sparsity) by driving several one-hot encoded employment/profession coefficients to zero. This is beneficial when many features are irrelevant. Here, the zeroed features are low-magnitude one-hot categories, so accuracy is preserved while the model becomes simpler.
- **Elastic Net:** With l_1-ratio $= 0.8$ the model leans strongly towards Lasso behaviour, achieving similar sparsity with the stability benefit of the L_2 component for correlated feature groups. The result is nearly identical to Lasso in this case.

11 Observations and Conclusion

Observations

1. All four models achieve $R^2 \approx 0.935$ and RMSE $\approx$ 15,250 USD on the test set, confirming that the loan amount target has a strong linear relationship with the available features.

2. `Property Price` and `Current Loan Expenses (USD)` are consistently the top predictors across all models. `Income (USD)` and `Credit Score` also contribute meaningfully.

3. Regularization primarily serves to *simplify* the model (fewer active features via Lasso) rather than to improve accuracy, since the baseline linear model already generalises well.

4. Lasso achieves the best MAE (10,885 USD) and RMSE (15,255 USD) on the test set, marginally ahead of the other models, while also producing a sparser coefficient vector.

5. RandomizedSearchCV found equivalent hyperparameters to GridSearchCV for Elastic Net while evaluating far fewer combinations — a practical advantage for larger search spaces.

6. The learning curve shows RMSE stabilising by 8,000–10,000 training samples, so the full 16,000-sample training set is likely sufficient; adding more data would yield diminishing returns.

Conclusion

This experiment compared four linear regression variants on a real-world loan amount prediction task. The baseline Linear Regression model already achieves strong performance ($R^2 = 0.9356$), demonstrating that the relationship between applicant features and requested loan amounts is predominantly linear. Ridge and Elastic Net provide marginal accuracy improvements while reducing coefficient variance; Lasso goes further by zeroing out low-value one-hot features, producing the most interpretable and slightly best-performing model.

Recommended model: Lasso Regression with $\alpha = 10$ (tuned via 5-fold GridSearchCV). It matches or slightly exceeds the other models on all test metrics, uses fewer active features (simpler to maintain), and trains in 51 ms — a practical, interpretable, and efficient choice for production deployment.

12 References

[1] Chitraju Vishnu Vineeth, "Experiment 3 Regression Analysis With Linear and Regularized Models" GitHub Repository. [Online]. Available: https://github.com/VishnuVineeth14/ML_LAB_ASSIGNMENTS/blob/main/Assignment3/Experiment_3_Regression_Analysis. [Accessed: Jul. 2026].